

Introduction	1
Mathematical definition of Intelligence	2
Major definitions	2
1. Universal Intelligence (Legg & Hutter)	2
2. Intelligence as Expected Utility Maximisation	2
3. Intelligence as Bayesian Inference	3
4. Intelligence as Prediction	3
5. Intelligence as Compression	4
6. Intelligence as Information Gain	4
7. Intelligence as Optimisation	5
8. Intelligence as Search	5
9. Intelligence as Approximation	5
10. Intelligence as Representation Learning	6
11. Intelligence as World Modelling	7
12. Intelligence as Control	8
13. Intelligence as Generalisation	8
14. Intelligence as Minimum Description Length	8
15. Intelligence as Free Energy Minimisation	9
A unified mathematical expression	9

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Introduction

I am working on an online course / book called - [ONLINE COURSE - MATHEMATICAL FOUNDATIONS OF ARTIFICIAL INTELLIGENCE](#)

I like to express this idea through three questions

- 1) Is there a mathematical way to define intelligence?
- 2) If current LLM models are simply prediction next sequences (sentences) - why are they so effective especially across domains? In other words whats the connection between sentence predication and writing code?
- 3) What does the success of reasoning loops tell us about intelligence?

I will address the first of these questions here - ***Is there a mathematical way to define intelligence?*** The short answer is no - but its a great [thought experiment Gedankenexperiment](#) to understand the maths of AI as we see below

Mathematical definition of Intelligence

There is no single universally accepted mathematical definition of intelligence, but several influential frameworks express it mathematically. Interestingly, they capture different aspects of intelligence: for example, prediction, adaptation, optimization, compression, reasoning, and decision making.

Below are the major mathematical formulations.

Major definitions

1. Universal Intelligence (Legg & Hutter)

Perhaps the closest thing we have to a mathematical definition is a collection of definitions . - [A Collection of Definitions of Intelligence](#) by [Legg and Hutter](#) along with their own definition ([Legg and Hutter Intelligence definition](#))

Definition: An intelligent agent performs well across many environments. The intelligence of an agent equals **sum over all environments** weighted by **how simple the environment is** times **how much reward the agent achieves**.

This is probably the most elegant mathematical definition of intelligence.

$$\xi(\pi) = \sum_{\mu} 2^{-K(\mu)} V^{\pi}_{\mu}$$

π = agent

μ = environment

$K(\mu)$ = [Kolmogorov complexity](#)

V^{π}_{μ} = expected reward.

Intelligence is the weighted sum of expected performance across environments.

2. Intelligence as Expected Utility Maximisation

An intelligent system chooses actions that maximise future utility.

$$U = E[\sum \gamma^t r_t]$$

where

- r_t = reward
- γ = discount factor
- E = expectation

This is the basis of Reinforcement Learning.

Intelligence becomes **finding actions that maximise expected cumulative reward**.

3. Intelligence as Bayesian Inference

An intelligent system continually updates its beliefs based on the familiar Bayes rules.

$$P(A \mid B) = (P(B \mid A) \times P(A)) / P(B)$$

where

- H = hypothesis
- D = observed data

Learning is simply

Prior - Evidence - Posterior

In this case, Intelligence becomes **efficient and continuous belief updating**.

4. Intelligence as Prediction

One view (especially in modern deep learning) is that intelligence is prediction.

Mathematically,

$$\hat{y} = f(x)$$

where

- x = input
- f = learned model
- $\hat{y}$ = prediction

Learning means finding a function f that minimises prediction error. For example Mean Squared Error or Cross Entropy

5. Intelligence as Compression

An intelligent system discovers regularities. The better it compresses data, the better it understands.

Kolmogorov Complexity $K(x)$ is the length of the shortest program that generates x. Intelligence becomes finding short descriptions for complicated observations.

This links intelligence with

- abstraction
- representation
- generalisation

6. Intelligence as Information Gain

Entropy and Information Gain are key concepts used in models such as Decision Trees (machine learning) to decide which feature should be used to split the data.

Entropy measures the uncertainty or impurity in a dataset. High entropy → data is mixed (more uncertainty). Low entropy → data is pure (less uncertainty).

Information Gain: Information Gain (IG) measures how much uncertainty is reduced after splitting the dataset on an attribute. Information Gain tells us: "How much information do we gain by knowing the value of this feature?" The attribute with the highest information gain is chosen for the split.

We can consider the relationship Between Entropy and Information Gain as follows: Entropy = measures impurity before or after a split; Information Gain = reduction in entropy caused by a split. Hence, Information Gain = Entropy Before Split – Entropy After Split

Decision Trees Choose the feature with the highest information gain. Think of entropy as "disorder" and information gain as "how much disorder is removed by asking a question." An intelligent learner reduces uncertainty by choosing the feature with the highest information gain.

This can be represented as

$$H(X) = -\sum_x p(x) \log_2 p(x)$$

where:

- **H(X)** = entropy (uncertainty in the random variable X)
- **p(x)** = probability of outcome x
- **log₂** = logarithm base 2 (entropy measured in bits)

$$IG = H(\text{before}) - H(\text{after})$$

7. Intelligence as Optimisation

Almost every AI system solves $\min L(\theta)$ where θ = parameters L = loss function. Intelligence becomes finding parameter values that minimise loss.

8. Intelligence as Search

An intelligent system searches a huge hypothesis space.

Search Problem **$\operatorname{argmax}_x f(x)$**

Examples: chess, theorem proving, planning, robotics

Many reasoning systems can be expressed this way.

$\operatorname{argmax}_x f(x)$ means: *"Find the value of x that makes the function $f(x)$ as large as possible."*
Notice that it returns the input, not the maximum value itself.

Here, arg = argument (the input to the function); $\max$ = maximum; x = the variable we are allowed to choose; $f(x)$ = the function we want to maximize

So, it means, we are trying to find out "Which value of x gives the largest value of $f(x)$?"

In Machine Learning, Suppose a classifier predicts probabilities:

Class	Probability
-------	-------------

Cat	0.10
-----	------

Dog	0.75
-----	------

Bird	0.15
------	------

The model predicts Dog because Dog has the highest probability.

9. Intelligence as Approximation

Intelligence can be seen as the ability to build good (approximate) models.

Reality - Approximation - Decision

Mathematically

$$f(x) \approx \hat{f}(x)$$

Good intelligence constructs useful approximations and useful models approximate reality.

10. Intelligence as Representation Learning

Deep learning suggests intelligence depends on learning useful representations.

Instead of x learn $z = \phi(x)$ where z is a latent representation. Good representations separate important factors of variation. Deep learning discovers useful latent representations.

The idea of learning a latent representation instead of the original function is one of the central concepts of modern AI.

When we write $z = \phi(x)$

we are saying: Instead of learning directly from the raw input x , first transform it into a more useful internal representation z .

This internal representation is called a latent representation (or embedding, feature representation, or hidden representation, depending on the context).

Imagine you want to recognize faces.

The raw input is an image:

x = pixels A 224×224 RGB image contains $224 \times 224 \times 3 = 150,528$ numbers

The raw pixels themselves are not very meaningful. Instead, a neural network gradually transforms them from raw Pixels to Edges to Corners to Eyes, Nose, Shape of a face comprising eyes and nose to an Identity(person)

The intermediate representation is $z = \phi(x)$. Instead of working directly with pixels, the model learns a much richer representation. A latent representation is a compressed set of variables that captures the important structure of the data. For example, instead of 150,528 pixels the network may produce $z = 512$ numbers. Those 512 numbers are not manually designed features, the network learns them automatically.

Classically we might learn $y = f(x)$ where x = input, y = output. Modern deep learning instead learns $x \rightarrow z = \phi(x) \rightarrow y = g(z)$ or $y = g(\phi(x))$ where ϕ is the representation function g is the prediction function. For example, in a cat classifier, the numbers for the interim representation represent features of the cat. This decomposition is fundamental to representation learning.

Why learn from intermediate representations? Suppose we try to learn directly Pixels $\rightarrow$ Cat - this is difficult because pixels change with lighting, rotation, scale, camera, background. The representation is an abstraction of the salient features and thus can be used to detect many variants of the same object(ex cat).

We see the same with Autoencoders. The flow is Image - Encoder - Latent vector z - Decoder - Reconstructed image.

The encoder computes $z = \phi(x)$. The decoder tries to reconstruct x . If reconstruction is good, z has captured the essential information.

We see the same idea with Large Language Models. For example, if the input is "The cat sat on the mat.". The Tokenizer creates the Embeddings. The Transformer layers create the Hidden states. Those hidden states are the latent representation. The model never reasons directly over raw words. It reasons over learned vectors. In this sense, latent means hidden but inferred.

The geometry of the latent space reflects semantic similarity. A famous example is: Suppose word embeddings learn $\text{King} - \text{Man} + \text{Woman} \approx \text{Queen}$. This works because King, Queen, Man, Woman live in a latent space where relationships become approximately linear. The latent representation has captured meaningful semantic structure without being explicitly programmed.

This idea can be generalized. Representation learning is one of the biggest shifts in AI over the past decade has been from hand-engineered features to learned representations.

11. Intelligence as World Modelling

Many researchers argue intelligence requires an internal model.

The agent learns

$$P(s_{t+1} \mid s_t, a_t)$$

where

- s = state
- a = action

Then plans using that model.

This is central to

- robotics
- cognitive science
- model-based reinforcement learning

The probability of the next state (s_{t+1}), given the current state (s_t) and the current action (a_t). This is the fundamental equation behind world models, Markov Decision Processes (MDPs), reinforcement learning, and model-based planning.

12. Intelligence as Control

Control theory views intelligence as stabilising systems. The intelligent system chooses actions

to keep the system near desired states.

$$\mathbf{x}[k+1] = \mathbf{A} \mathbf{x}[k] + \mathbf{B} \mathbf{u}[k]$$

The next state of the system is determined by:

- $\mathbf{x}[k]$ — the current state
- $\mathbf{A}$ — how the system naturally evolves
- $\mathbf{u}[k]$ — the control input
- $\mathbf{B}$ — how the control input affects the system

The future state equals the natural dynamics of the system plus the effect of the chosen control action.

13. Intelligence as Generalisation

Perhaps the single most important mathematical concept. An intelligent system performs well on unseen data. This is arguably more important than memorisation.

Perhaps the single most important mathematical concept in artificial intelligence. An intelligent system performs well on **unseen data**, not just on the examples it has already encountered. Intelligence is therefore measured by the ability to **generalise**, rather than to memorise.

The mathematical goal is to learn a function that performs well on new observations drawn from the same underlying distribution.

Training teaches a model from examples. Generalisation determines whether that learning transfers to new situations. An intelligent system does not simply remember the past—it applies what it has learned to the future. Without generalisation, the model memorises. With good generalisation, the model adapts to unseen situations, learning transfers across tasks

14. Intelligence as Minimum Description Length

Closely related to [Occam's Razor](#), choose the model that minimises $L(\text{Model}) + L(\text{Data} \mid \text{Model})$. This balances simplicity and predictive accuracy

15. Intelligence as Free Energy Minimisation

A prominent theory in neuroscience, associated with [Karl Friston](#). The brain continually updates internal models to minimise surprise. One of the most influential mathematical theories of intelligence proposes that the brain is fundamentally a **prediction machine**. Rather than passively observing the world, it continually predicts what it expects to perceive and compares those predictions with reality. When predictions are wrong, the brain updates its internal model

to reduce future errors. The mathematical objective is to minimise **surprise**. These predictions are generated from an **internal model** of the world.

The central quantity is **prediction error**. $\text{Prediction Error} = \text{Observation} - \text{Prediction}$ The goal is to make this error as small as possible.. In this context, Surprise has a precise mathematical meaning. **According to the Free Energy Principle, intelligence emerges from continually improving an internal model of the world by minimising prediction error and reducing surprise.**

This idea is somewhat similar to Bayesian - but with some key differences. Bayes tells us **how to update beliefs** when new evidence arrives. Friston's theory says the brain is constantly updating beliefs. But the Free Energy Principle goes one step further. Instead of saying "Update your beliefs." - it says "Act so that your future observations become less surprising." Now action enters the picture. Instead of merely updating beliefs, the agent may also change the world. In contrast, Bayes only updated beliefs.

A unified mathematical expression

Although no single equation captures all aspects of intelligence, we can write a conceptual formulation that combines the main ideas:

Intelligence $\approx$ Learning + Representation + Prediction + Reasoning + Planning + Adaptation + Generalisation

In this sense, we can say: ***Intelligence is the capability of an agent to construct increasingly useful internal representations of the world, use them to predict, reason, plan, and act, and continually improve those representations from experience so as to maximise long-term goal achievement across diverse environments.***

This synthesis brings together the major mathematical themes:

- **Learning** → Bayesian inference, statistical estimation
- **Representation** → latent variables, manifolds, embeddings
- **Prediction** → probability theory and function approximation
- **Reasoning** → search, logic, inference, optimization
- **Planning** → control theory and reinforcement learning
- **Generalisation** → statistical learning theory
- **Adaptation** → information theory and continual learning

Taken together, these perspectives suggest that intelligence is not a single algorithm but an emergent property arising from the interaction of learning, representation, computation, optimisation, and decision-making under uncertainty.

Another way to think of Intelligence is via [David Marr's](#) framework which explains **how any intelligent system should be analyzed**. Marr does **not** ask: "*What is intelligence?*" Instead, he asks: "*How should we explain an information-processing system?*" His answer is that any cognitive system must be understood simultaneously at **three levels**.

Marr's Three Levels are:

- Computational: An objective - Examples: maximize reward, minimize prediction error
- Algorithmic: A function: Examples transformer, Bayesian inference, planning, search, reinforcement learning
- Implementation: A physical system: Examples brain, GPU cluster, robot

If you want to be part of my course please see [ONLINE COURSE - MATHEMATICAL FOUNDATIONS OF ARTIFICIAL INTELLIGENCE](#) or contact me.